TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Matthias Trenn	**Email**	matthias.trenn@ilt.fraunhofer.de
Institute Name	Fraunhofer Institute for Laser Technology ILT			
Address	Steinbachstraße 15, 52074 Aachen, Germany			

2. Technology Overview

Title of Proposed Technology	*Boosting electrolysis and fuel cell performance by increasing the surface area through laser structuring*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*New & Renewable Energy*	*Fuel Cells*
Concept of Proposed Technology	• *Surface modification by laser structuring* • *Increase surface area by factor of 1000* • *Generation of water-repellent surfaces for water management* • *These effects on laser-treated electrodes and BPP increase the performance of PEM electrolysis, alkaline electrolysis and fuel cells by 10-20%*			
Rationale for Joint R&D Project	• *As an expert in laser material processing technology and optical systems, Fraunhofer ILT can provide several high-power and ultrashort pulse laser systems as well as optical devices and machines for large-area processing.* • *Fraunhofer ILT has experience in the described field of laser processing of electrodes and BPP for hydrogen technology. This experience has been gained in several projects.*			

(max 10 pages)

	• *Fraunhofer ILT is looking for partners to optimize these processes specifically for industrial products, scaling the processes, and achieving market readiness.*
Key Words	1. *Electrolysis improvement* 2. Fuel cell performance increase 3. Surface enlargement 4. Water management 5. Laser structuring

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level ( )	End of the Project	Level ( )
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project*			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters*			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives*			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.*

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.*

6. References

(1)

(2)

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)